

UNIVERSITETET I OSLO

The perceptual motivation for the Norwegian merger of /ʃ/ and /ç/

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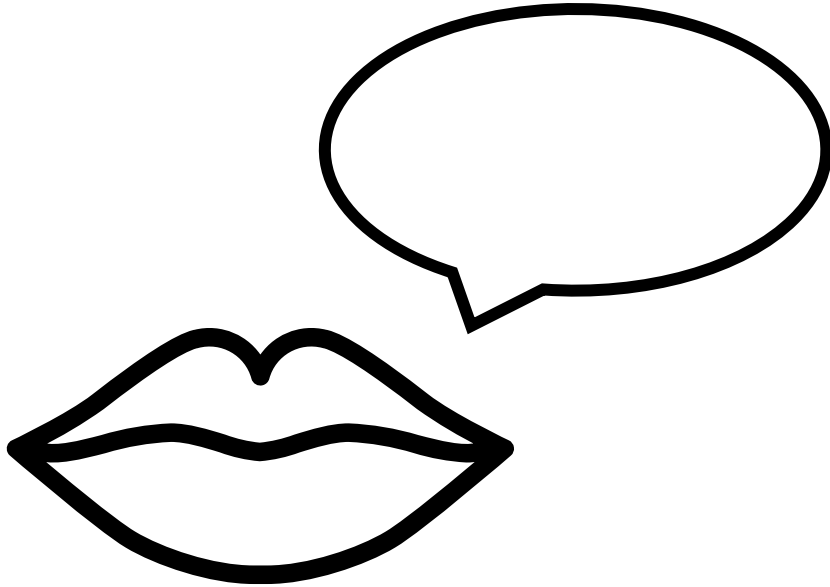
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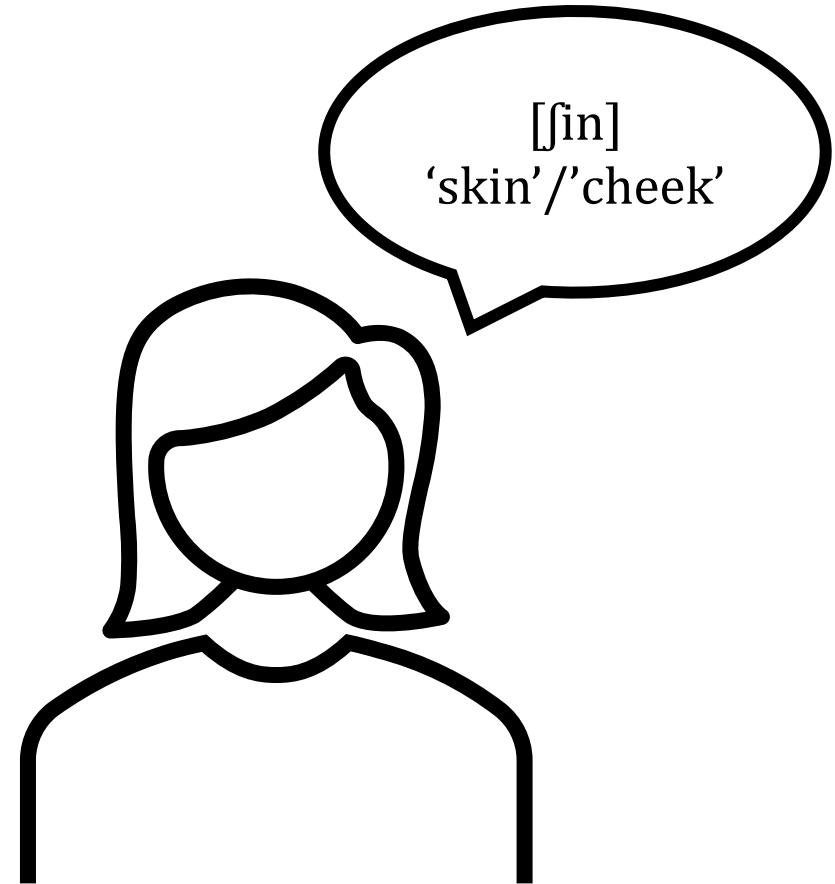
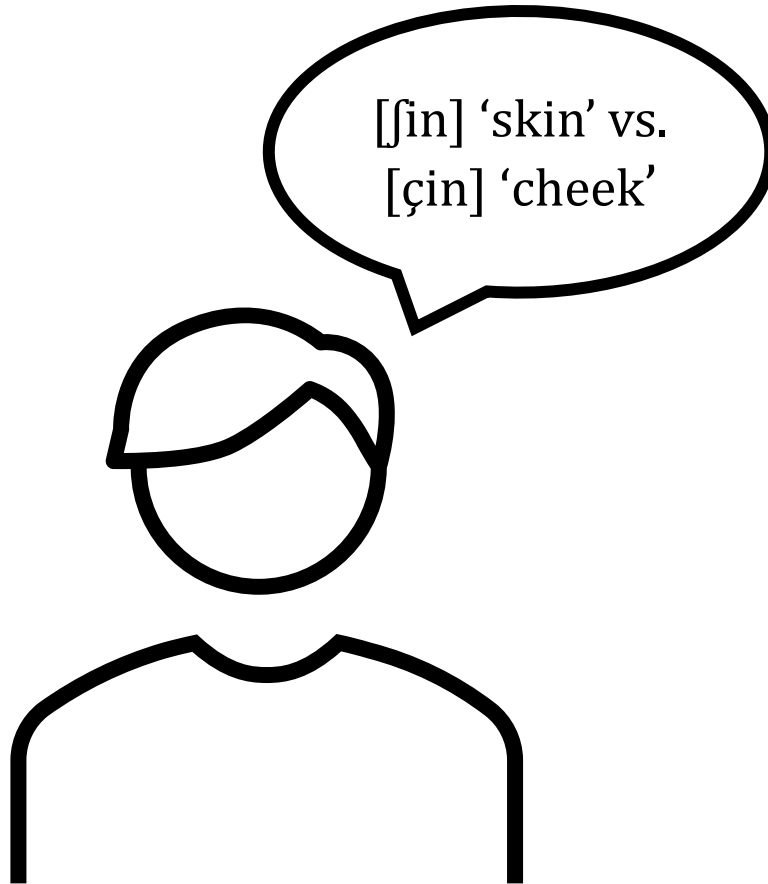
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What motivates sound change?



/ʃ/ and /ç/ in Norwegian



[Redacted]
Hvorfor sendes ikke lenger barn til logoped for feilaktig uttale av kj-lyden?

  133

275 kommentarer

 Liker

 Kommenter

 Del

[Redacted]
Har flere talefeil nå til dags, eller lærer ikke barn å snakke rent norsk? Det være seg av foreldre, og/eller på skolen? Jeg tenker spes. på kj-lyden, som har blitt til sh-lyd. Synes det er en skam at språket skal forringes på denne måten. 😞

   4

38 kommentarer

[Redacted]
Lingvister vil kalle dette språkutvikling, men i realiteten er det språklig latskap. Det er lettere å bruke én skj/sj/ki-lyd. Barn må bruke den forenklete lyden for ikke å høres ut som foreldrene sine, altså for å være kule.

2 u Liker Svar Del

1 

*Sjærlighet og sjole er kommet for å bli,
uansett hvor sint du blir*

“The merger has come to stay, no matter how
angry you get”

Helene Uri: Er det barnslig å
snakke om «sjærlighet»?

“Is it childish to have the merger?”

La kj-lyden dø i fred!

“Let the kj-sound die in peace!”

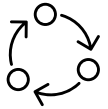
Du er ikke dum selv om du sier sjole

“You’re not stupid even though you have the merger”

The merger of /ʃ/ and /ç/



Found in various cities in Norway.



Descriptions of frequency and distribution.



No studies investigating the phonetic motivation.



[ç]: a difficult sound to produce?



Child language acquisition

Last phoneme acquired by children.



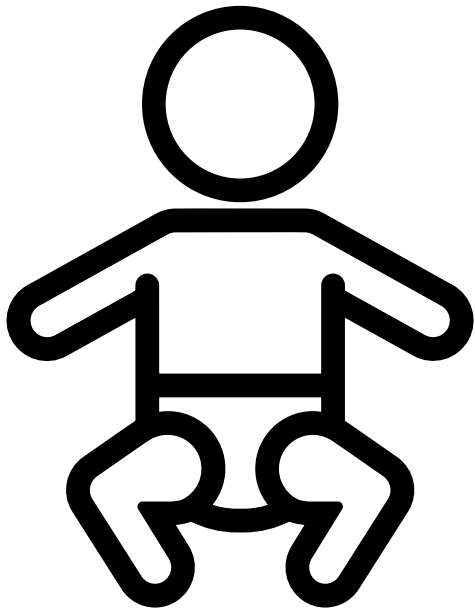
Cross-linguistic frequency

Rare sound cross-linguistically.



Articulation

Dorsal /ç/ more difficult than coronal /ʃ/.



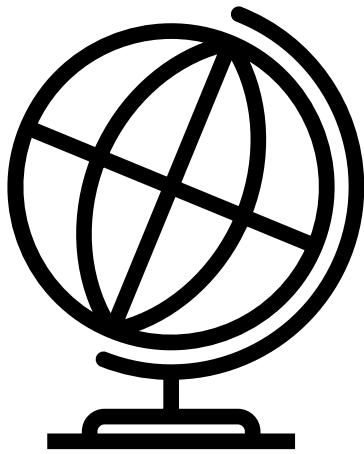
Child language acquisition

Vanvik (1971): one child speaker.

Fintoft et al. (1983): /ʃ ç s r/ all acquired late.

Simonsen (1990): high number of mistakes for both /ʃ/ and /ç/.

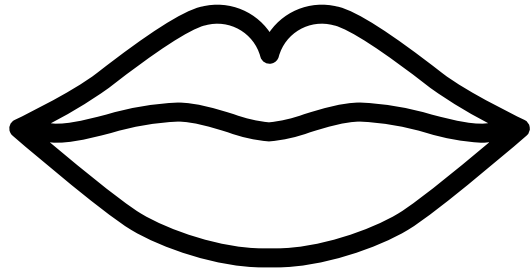
Croft (2000): child language not sufficient to explain sound change.



Cross-linguistic frequency

Maddieson (1984): rare phoneme.

Roach (2009) and Okada (1999): [ç] found as allophone in English and Japanese.



Articulation

Simonsen and Moen (2004): /ç/ is coronal for most speakers.

Evjen and Stausland (2025): outcome of the merger can be [ç].

Avoidance of [ç] cannot explain merging with [ʃ] in particular.

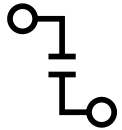
An alternative explanation



Potential acquisitional and typological patterns not necessarily due to articulation.



The motivation for the merger could be rooted in perception.



Problematic contrast between /ʃ/ and /ç/.



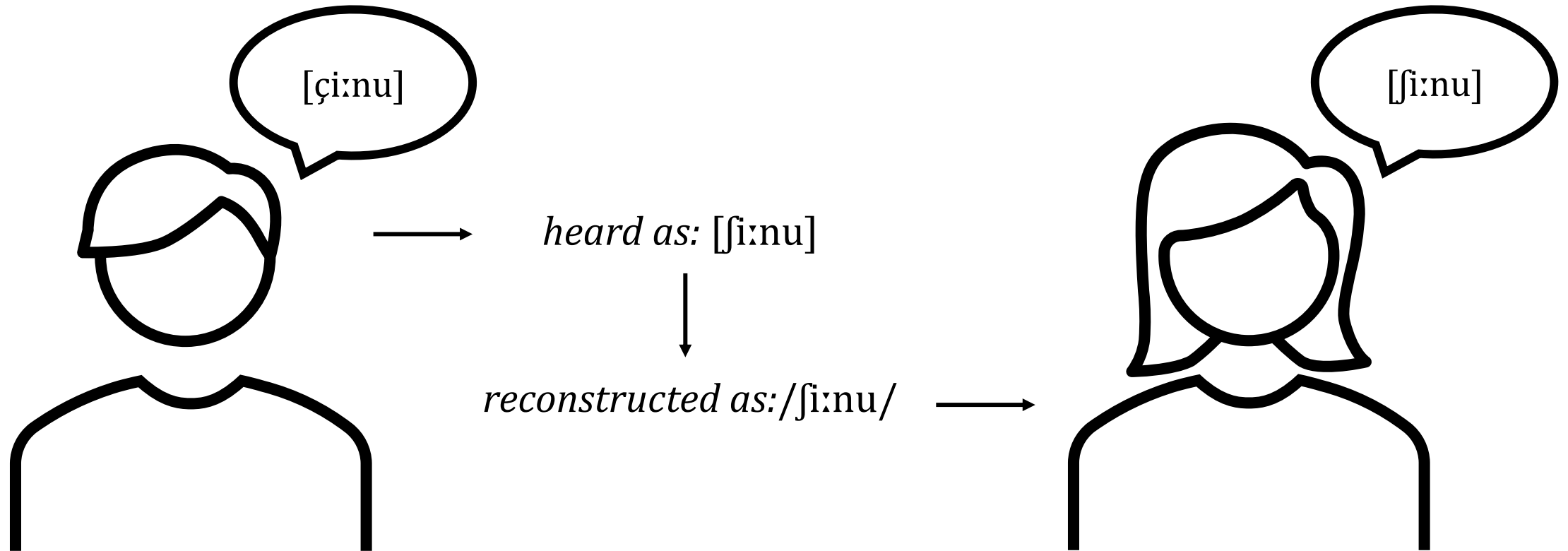
Perceptual similarity

Ohala (1981): listeners' misperception is a common source of sound change.

Blevins (2004): merger of /f/ and /θ/ in some English dialects.

Stausland Johnsen (2012): merger of apical alveolars and apical postalveolars in many Norwegian dialects.

Perceptual similarity



The current study



Perceptual distance between Norwegian voiceless fricatives: /f s ʃ ç/.

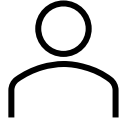


Confusability of /ʃ/ and /ç/ compared to other pairs.

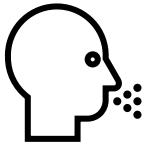


Smaller perceptual distance increases chance of misperceptions and subsequently merger.

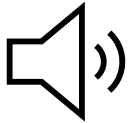
Methods



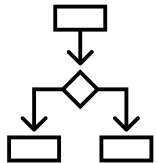
59 native Norwegian participants (45 female, mean age 24).



4 native Norwegian speakers (3 female, mean age 25).



24 CV nonce words per speaker (/f s ʃ ç/ × /i a u/ × 2 repetitions).



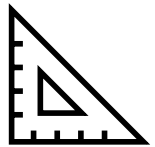
AX discrimination experiment.

Analysis



Six pairs: $\int - \varsigma$, $\varsigma - f$, $\varsigma - s$, $\int - f$, $\int - s$, $f - s$.

E.g.: $/\int - \int/$, $/\varsigma - \varsigma/$, $/\int - \varsigma/$, and $/\varsigma - \int/$.



Measures:

Proportions correct: rate of correct ‘different’ and ‘same’ trials.

d' (d-prime): $Z(\text{Hit rate}) - Z(\text{False alarm rate})$.

Analysis



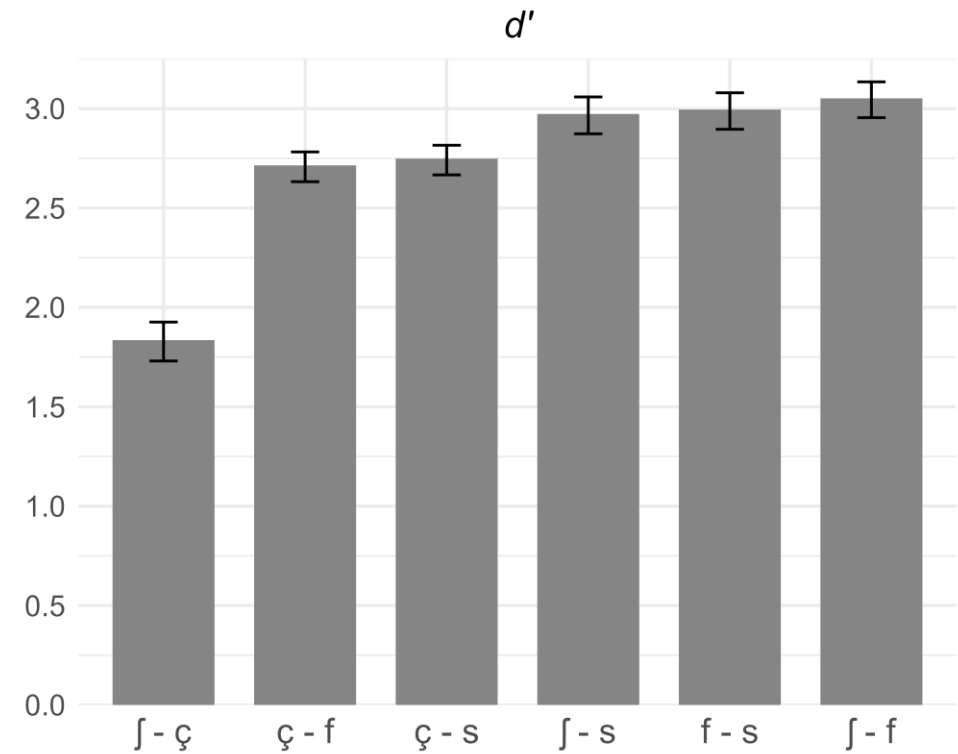
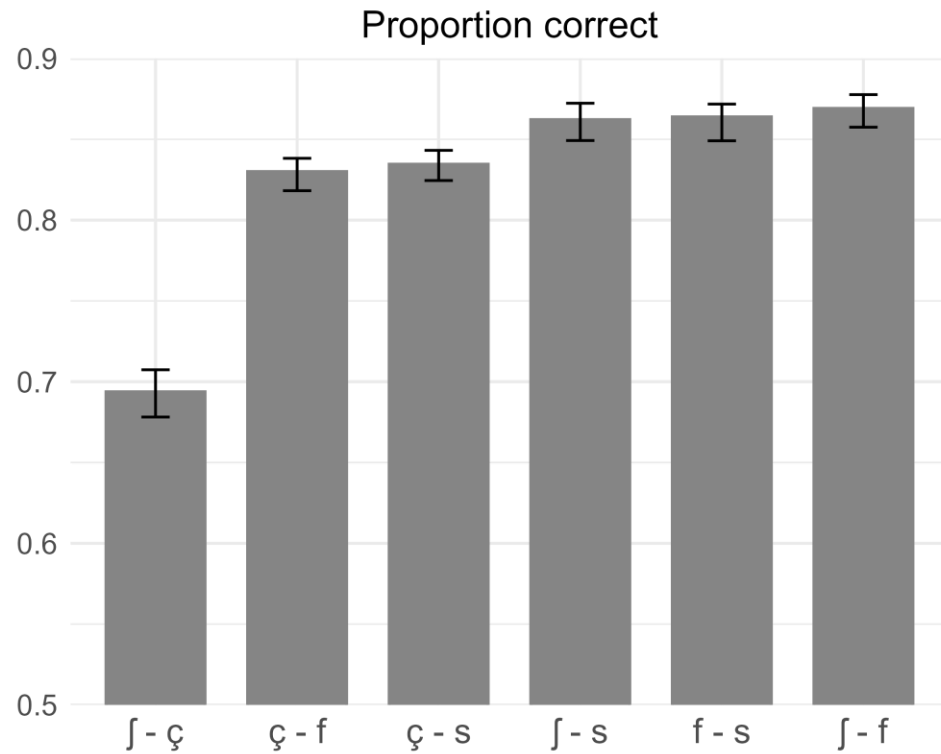
Statistics:

Multidimensional scaling (MDS): visual representation of perceptual distances as indicated by the d' values.

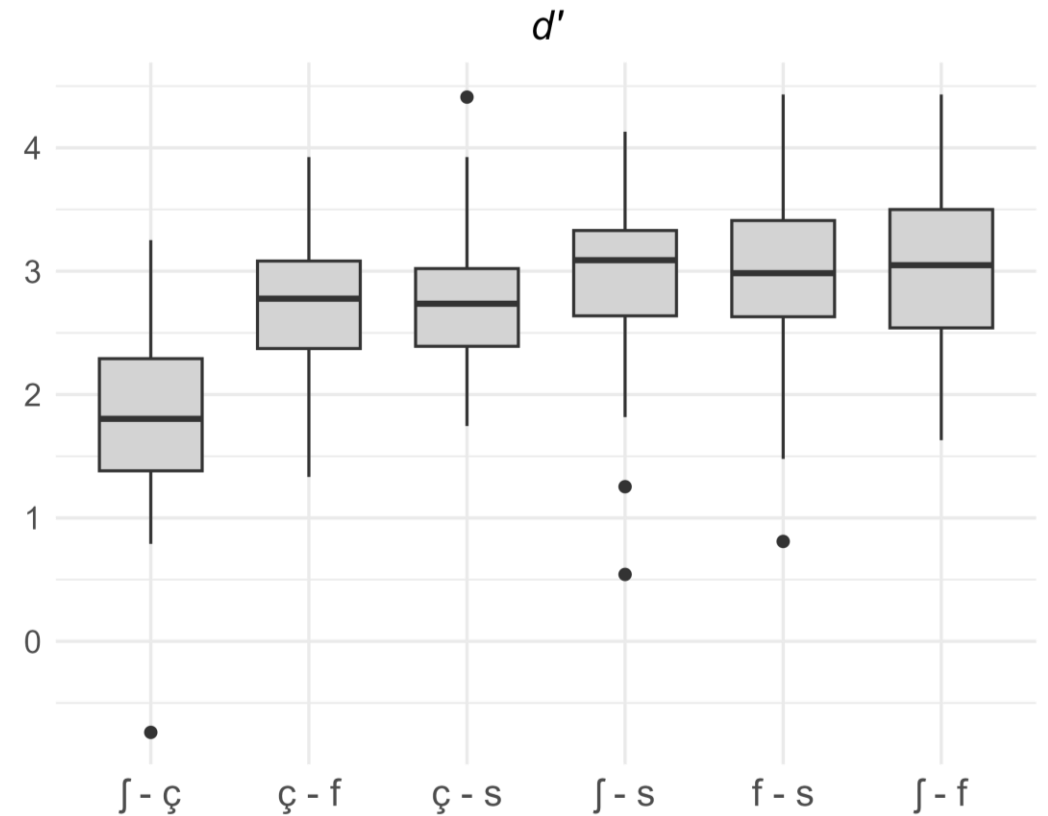
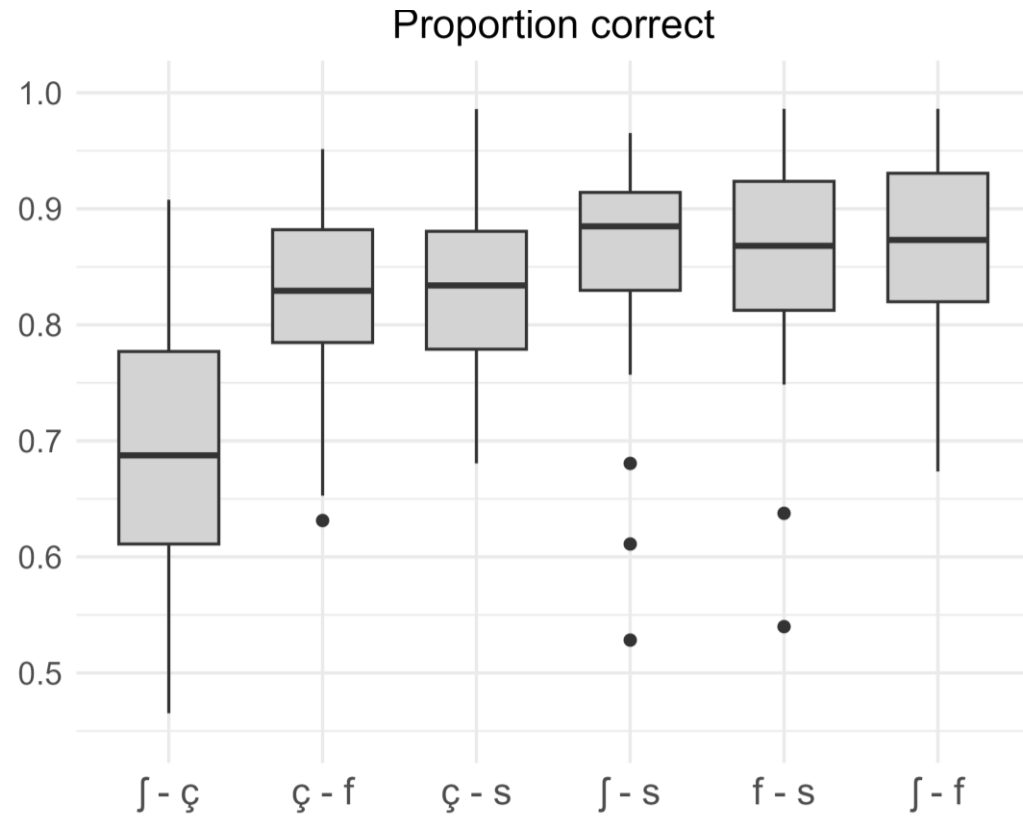
Bayesian mixed effects model: approximate posterior distributions of plausible parameter values, given the data, the model, and the priors.

$$d' \sim \text{pair} + (1 + \text{pair} \mid \text{listener}) + (1 + \text{pair} \mid \text{speaker})$$

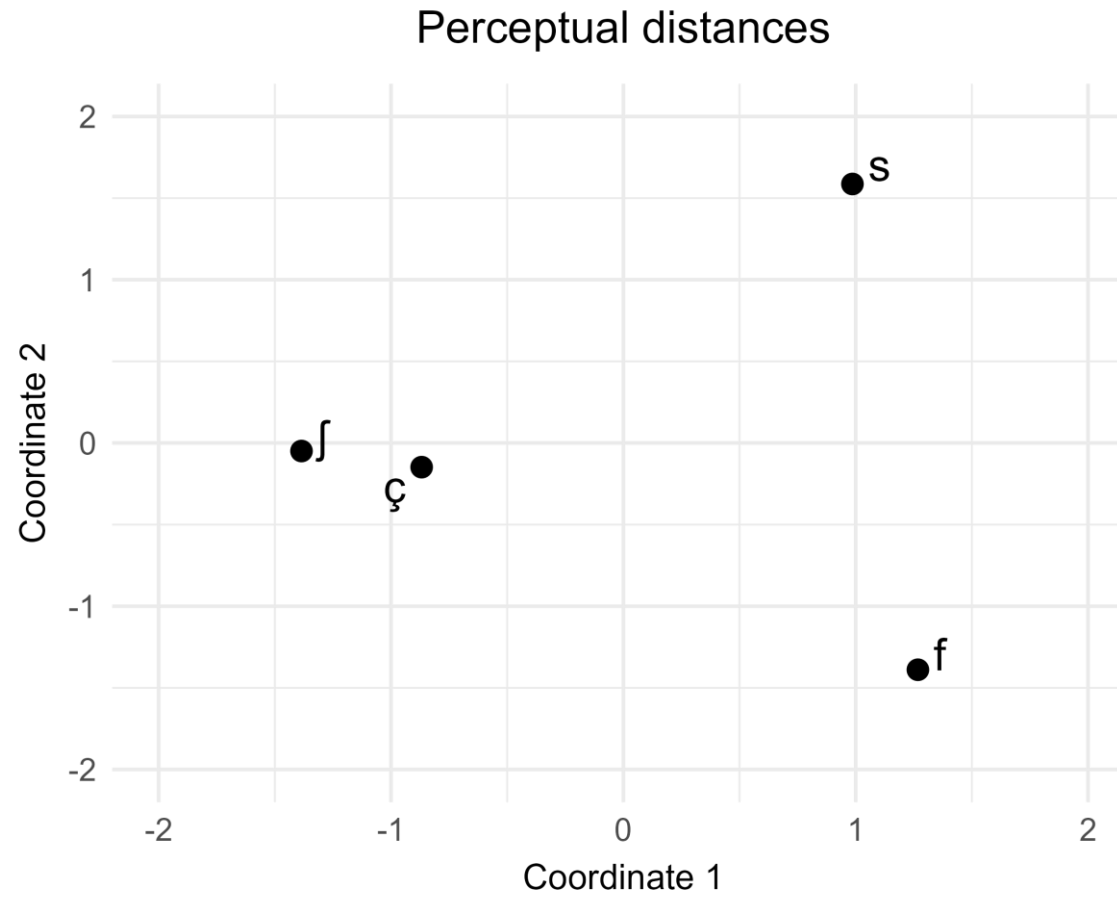
Descriptive results



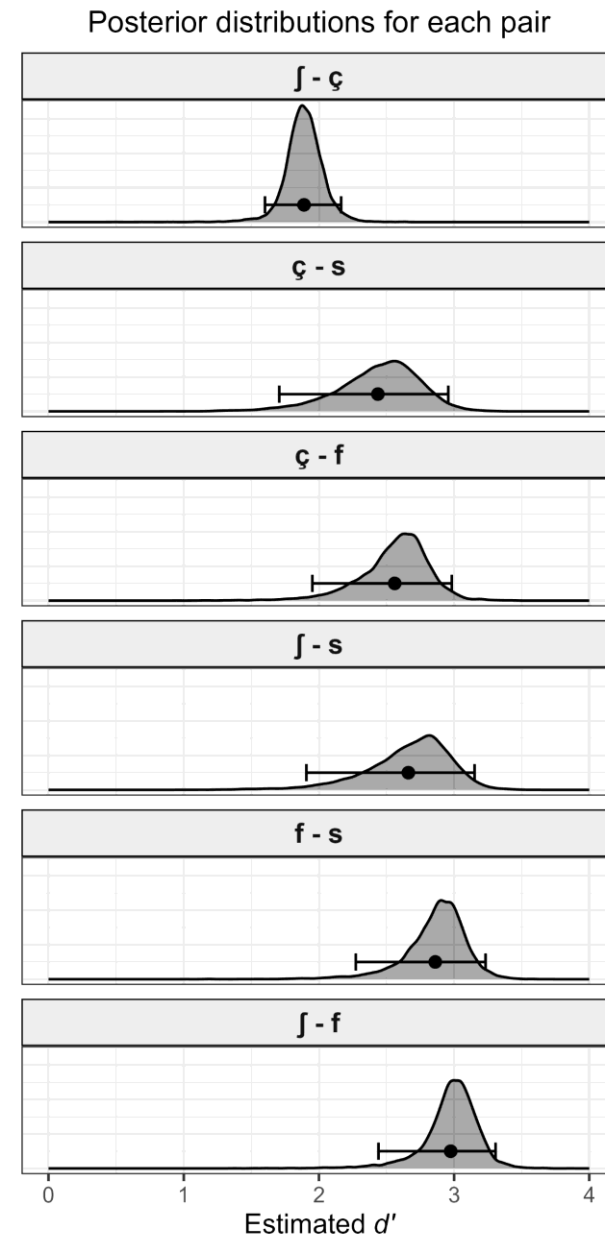
Descriptive results



Multidimensional scaling

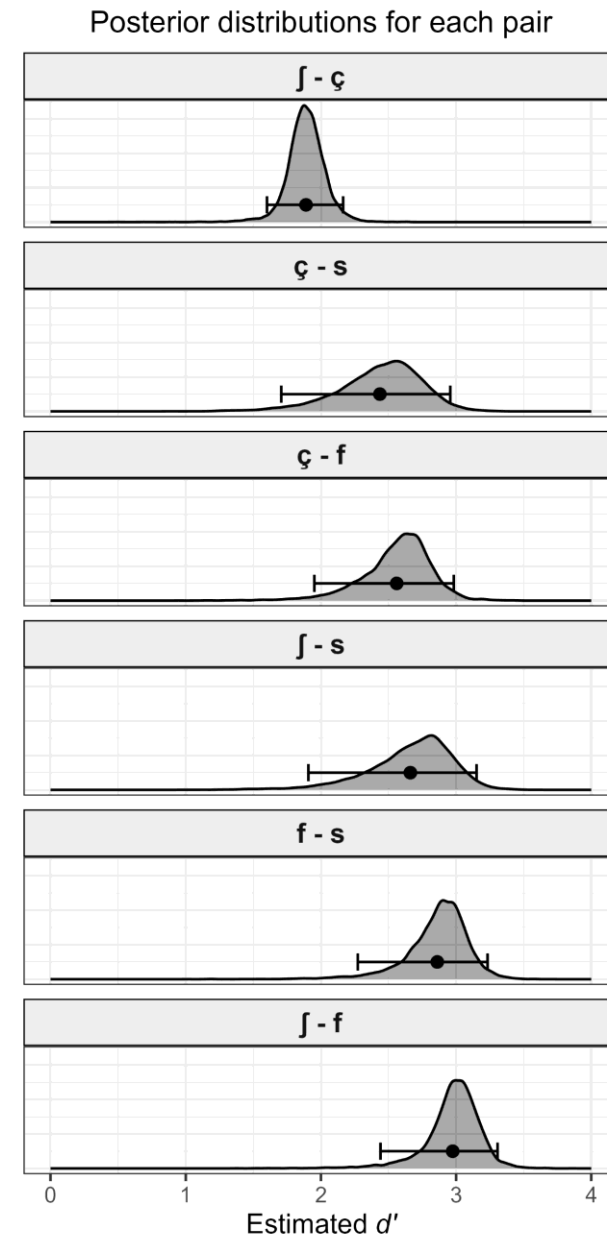


Regression model



Regression model

Pair	β	LCI	HCI	$\Pr(\beta > 0)$
$\int - \varsigma$	1.89	1.60	2.16	1
$\varsigma - s$	0.55	-0.12	1.03	0.953
$\varsigma - f$	0.67	0.10	1.03	0.986
$\int - s$	0.77	0.05	1.22	0.980
$f - s$	0.97	0.44	1.26	0.996
$\int - f$	1.08	0.63	1.33	0.998



Summing up



/ʃ/ and /ç/ were confused more often than other pairs.



Smaller perceptual distance between /ʃ/ and /ç/.



Listener misperceptions as explanation for the merger of /ʃ/ and /ç/.

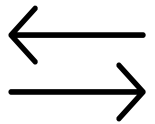
Summing up



First study investigating the motivation for the merger of /ʃ/ and /ç/.



Findings align with observed outcomes of the merger (Evjen and Stausland, 2025).



Important role of perception in driving forward sound change.

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